

Unit 2

Belief: Something you think is correct. Ex “ghosts are real”

Fact: Something that is true. Ex “ $7+1=8$ ”

Hypothesis: A testable guess. Ex “I think that if I punch a wall ill be sad”

Law: Explains how something happens, generally has math. Ex “ $F=ma$ ”

Theory: Robust culmination of years of work, cant currently be disproved. “Evolution, Tectonics

Doppler Effect

The **apparent** change in frequency of a sound wave from a **moving** sound source hear by a **stationary** observer.

Freq high = blue

Freq low = red



Big Bang Theory

It is the origin of the universe, there is no time space or even matter before it happened.

Singularity: Infinity hot and dense “no matter was formed because it was too hot”

Three points of evidence

1. red shift/universal spreading out “universal expansion”
2. cosmic microwave background radiation
3. Abundance of light elements: hydrogen and helium are vastly the most abundant element in the universe.

Red shift/Blue shift

Line spectra- “identity” of a burning element. Each element has its own unique spectra. Seen with a spectroscope.

How it works

1. Composition “Uniqueness of the absorption lines
2. Velocity = more shift/speed
3. Direction: rs/bs
4. Towards is blue
5. Away is red

